Data Structures

12. Binary Search Tree

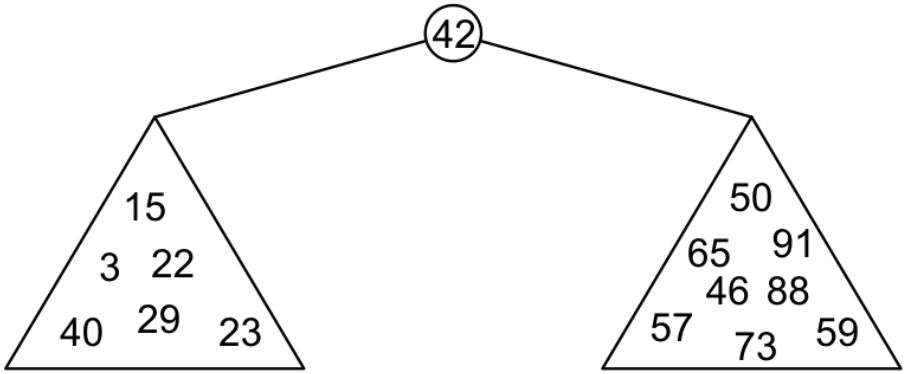
Binary Search Tree (BST)

- With a binary tree, we can dictate an order on the two children

- Binary Search Tree (BST) defines the following order:

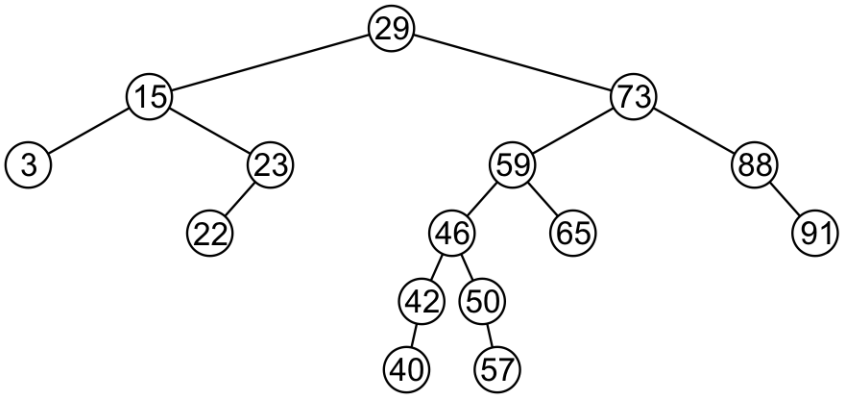
  – All elements in the left sub-tree to be less than the element stored in the root node, and

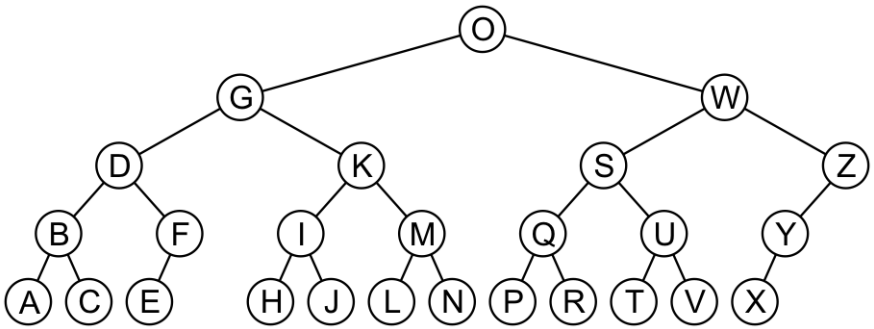
  – All elements in the right sub-tree to be greater than the element in the root object

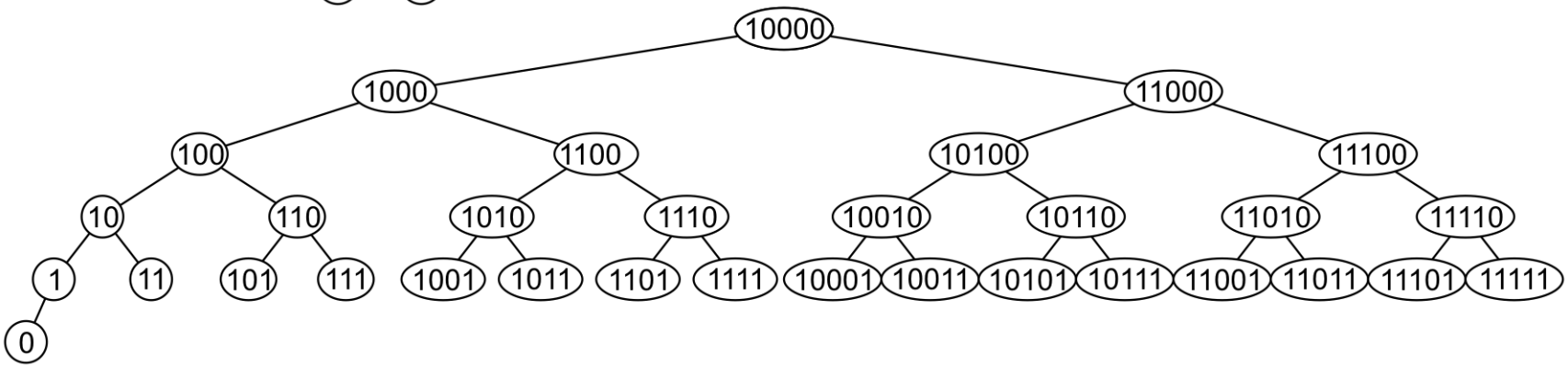


subtrees will themselves be binary search trees

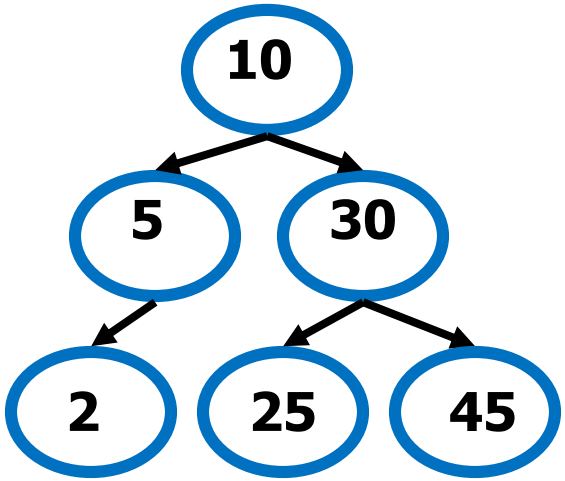
Binary Search Tree (BST) – Example (1)



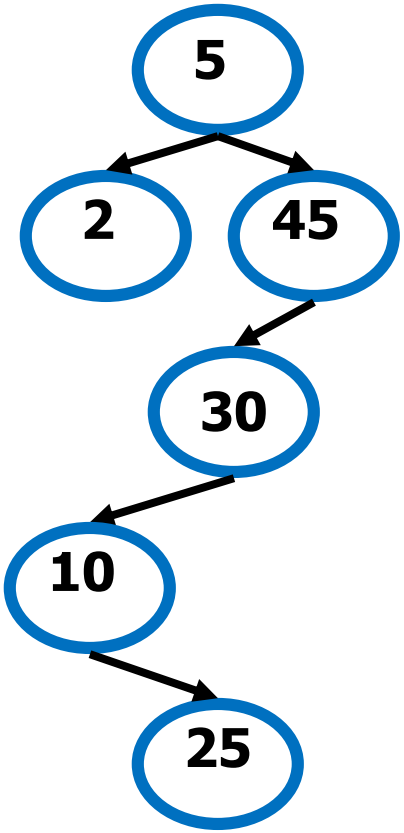




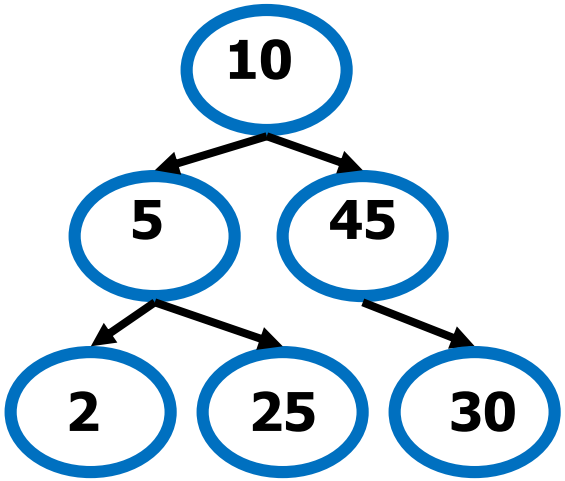
Binary Search Tree (BST) – Example (2)



BST



BST



Not BST

BST Operations

- Many operations one can perform on a binary search tree
  - Creating a binary search tree
  - Finding a node in a binary search tree
  - Inserting a node into a binary search tree
  - Deleting a node in a binary search tree
  - Traversing a binary search tree
- In the following, we will examine the algorithms and examples for all of the above operations

Creating BST

- A simple class that implements a binary tree to store integer values
  - A class called `IntBinaryTree`

- Node of binary search tree

```
struct TreeNode
{
     int value;
     TreeNode *left;
     TreeNode *right;
};
```

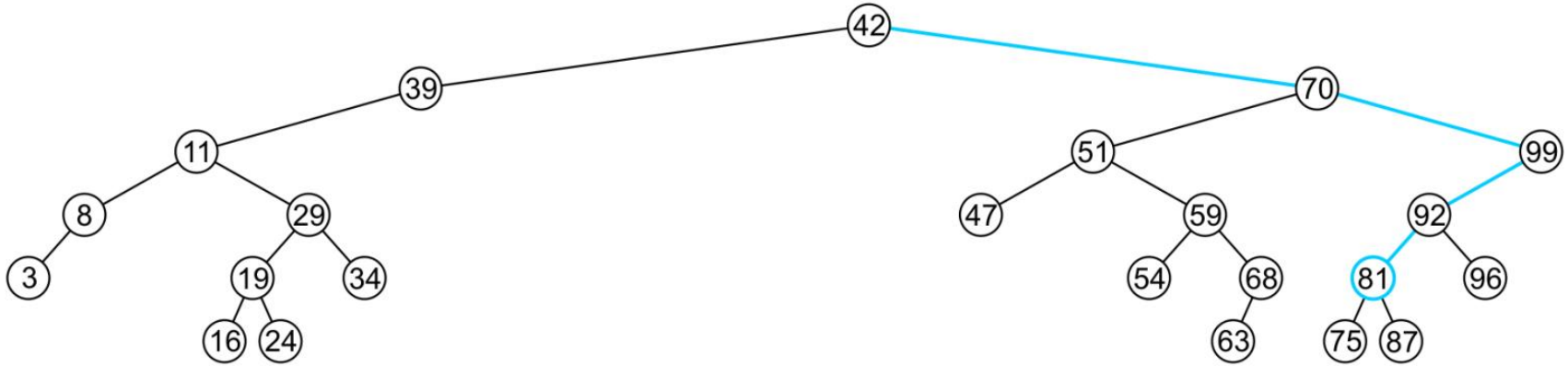
Creating BST – Class Definition

```
class IntBinaryTree {
   private:
      TreeNode *root;  // Pointer to the root of BST

      void destroySubTree(TreeNode *); //Recursively delete all tree nodes
      void deleteNode(int, TreeNode *&);
      void makeDeletion(TreeNode *&);
      void displayInOrder(TreeNode *);
      void displayPreOrder(TreeNode *);
      void displayPostOrder(TreeNode *);

   public:
      IntBinaryTree()              { root = NULL; }
      ~IntBinaryTree()             { destroySubTree(root); }
      void insertNode(int);
      bool find(int);
      void remove(int);            { deleteNode( num, root); }
      void showNodesInOrder()      { displayInOrder(root);   }
      void showNodesPreOrder()     { displayPreOrder(root);  }
      void showNodesPostOrder()    { displayPostOrder(root); }
};
```

Finding a Node in BST

- Recall that a BST has the following key property (invariant):
  - Smaller values in left sub-tree
  - Larger values in right sub-tree

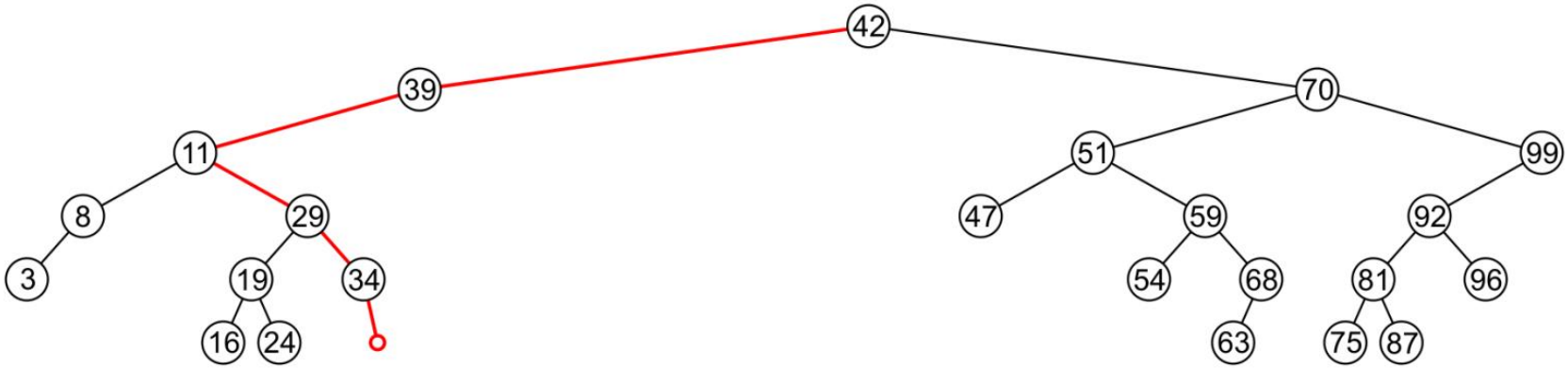
- For example: `find (81)`
  - Returns true if found



Finding a Node in BST

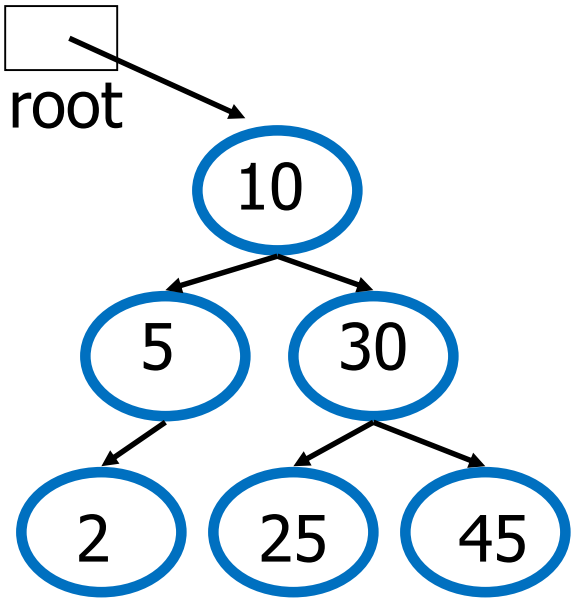
- Recall that a BST has the following key property (invariant):
  - Smaller values in left sub-tree
  - Larger values in right sub-tree

- For example: `find (36)`
  - Returns false if not found



Finding a Node in BST

- Example: `find(2)`



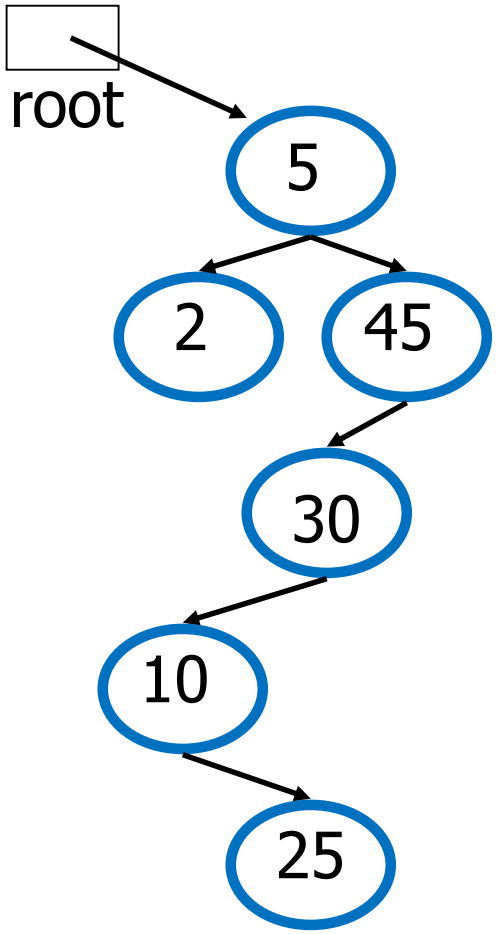
10 > 2, left

5 > 2, left

2 = 2, found

Finding a Node in BST

- Example: `find(25)`



5 < 25, right

45 > 25, left

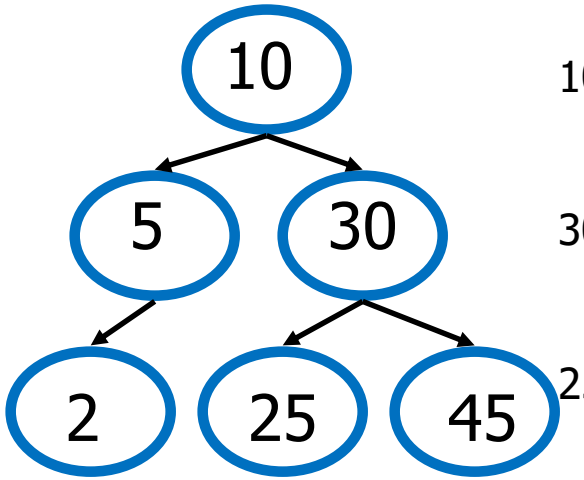
30 > 25, left

10 < 25, right

25 = 25, found

Finding a Node in BST – Implementation

```
bool IntBinaryTree::find(int num){
   // The function starts from the root
   TreeNode *nodePtr = root;

   while (nodePtr) {
      if (nodePtr->value == num)
         return true;  // value is found
      else if (num < nodePtr->value)
         nodePtr = nodePtr->left;
      else
         nodePtr = nodePtr->right;
   }
   return false; // value not found
}
```



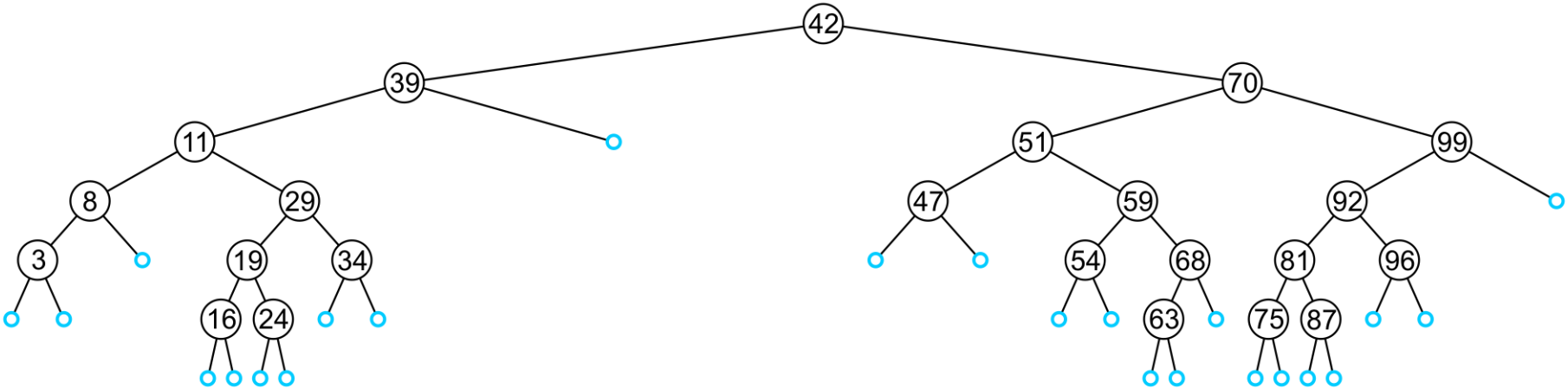
10 < 25, right

30 > 25, left

25 = 25, found

Inserting a Node in BST

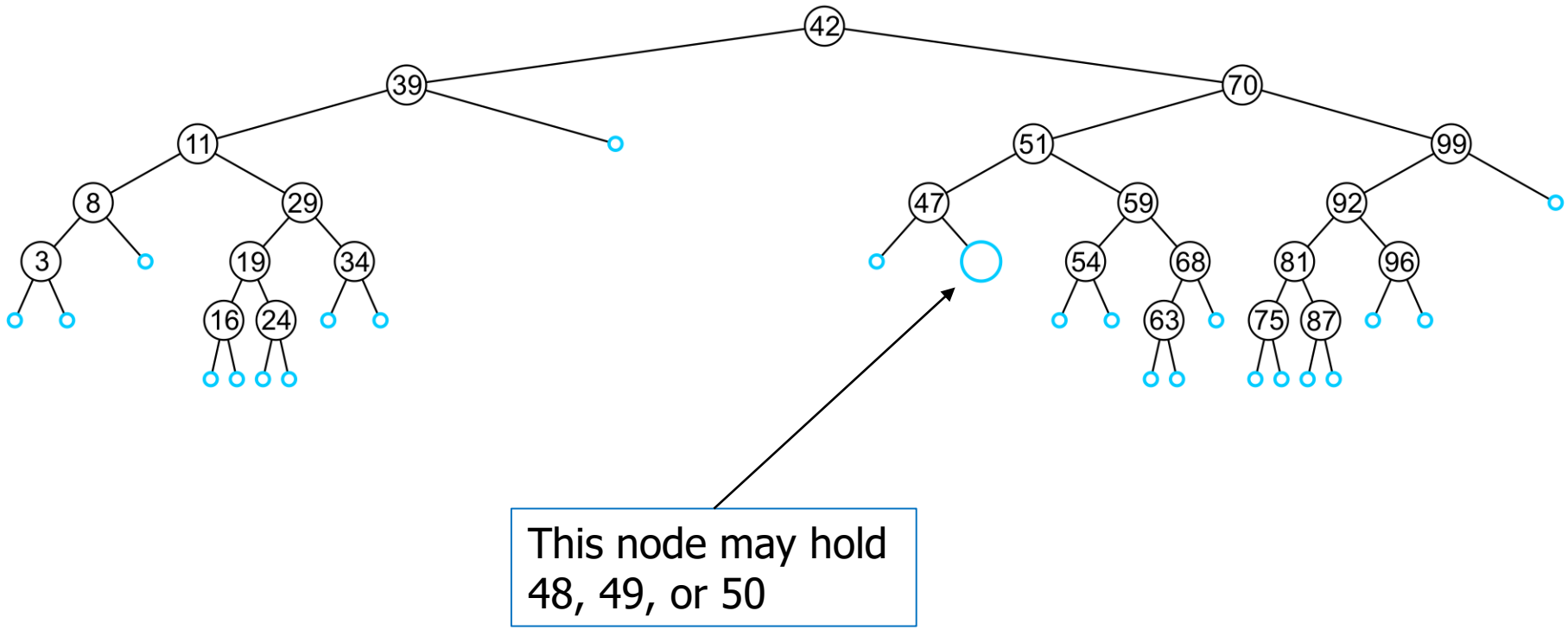
- An insertion will be performed at a leaf node
  - Any empty node is a possible location for an insertion



- Values which may be inserted at any empty node depend on the surrounding nodes

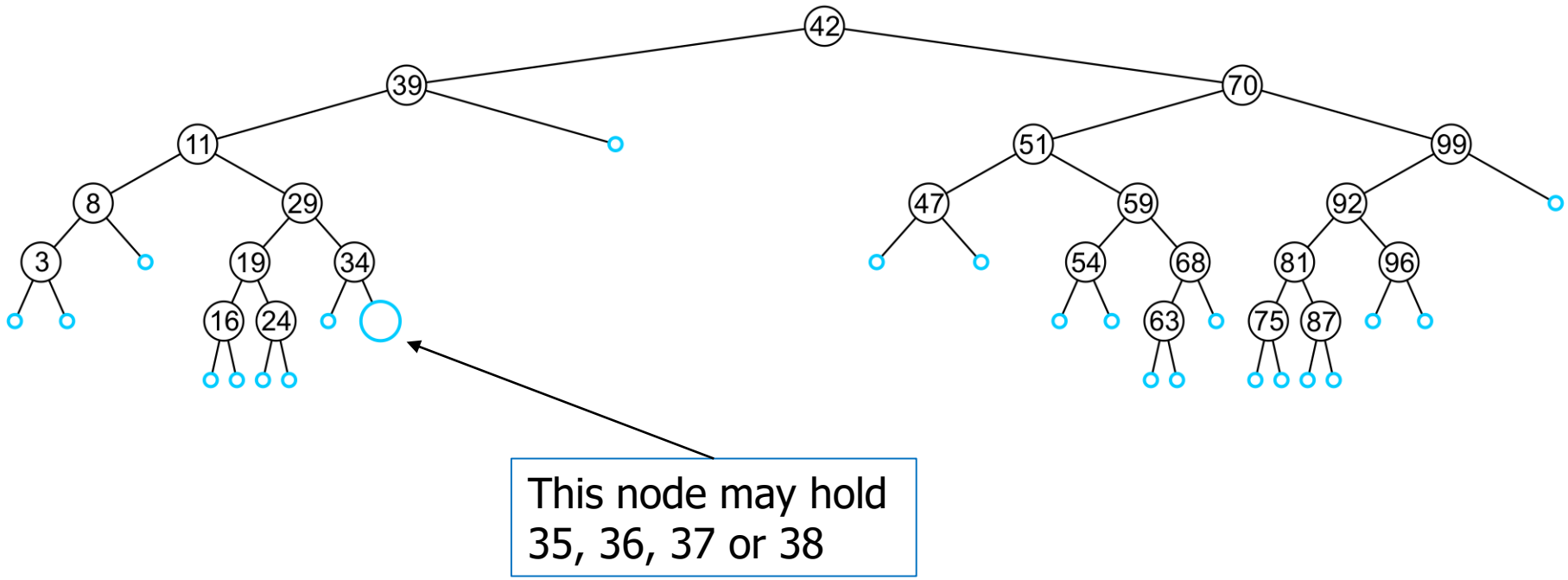
Inserting a Node in BST

- Which values can be held by empty node?



Inserting a Node in BST

- Which values can be held by empty node?



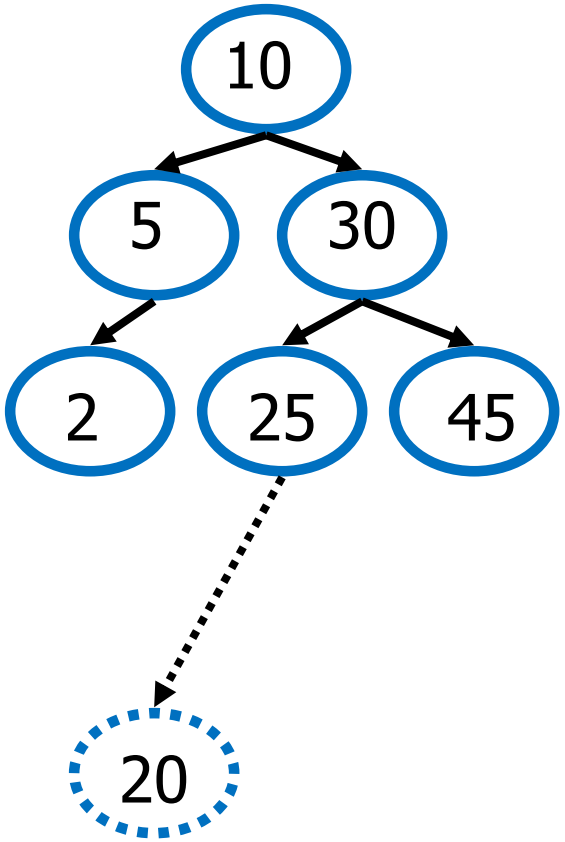
Inserting a Node in BST – Algorithm

- Like find, algorithm will step through the tree
  - If algorithm find the object already in the tree, it will return
    - The object is already in the binary search tree (no duplicates)
  - Otherwise, algorithm will arrive at an empty node
  - The object will be inserted into that location

- Why no duplicates?
  - In reality, it is seldom the case where duplicate elements in a BST must be stored as separate entities

Inserting a Node in BST – Example

- insertNode(20)



10 < 20, right

30 > 20, left

25 > 20, left

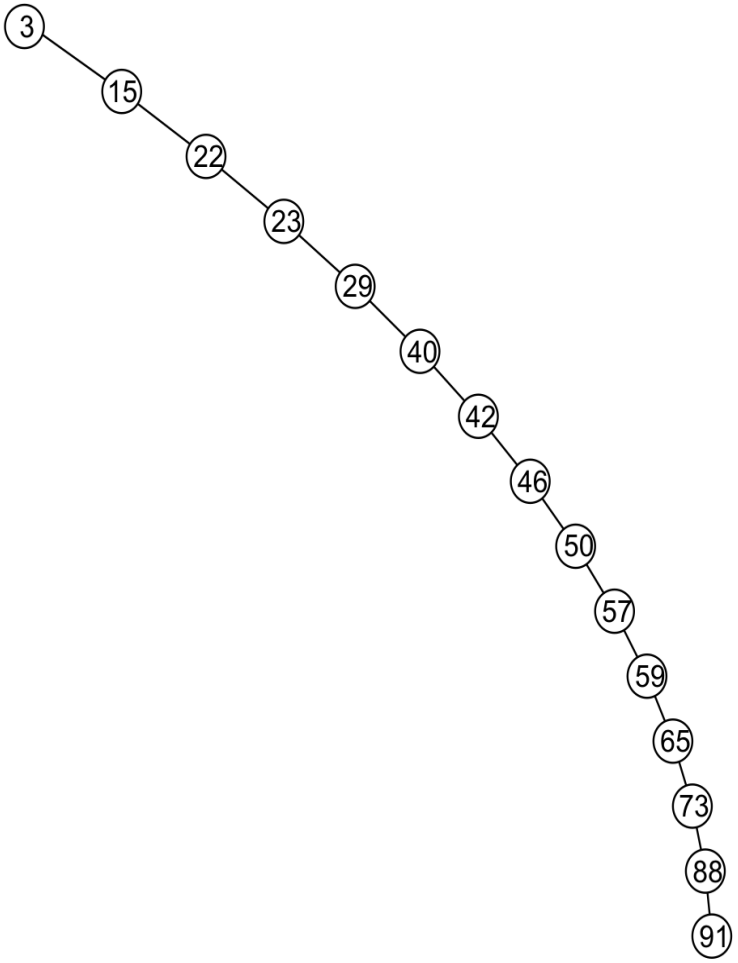
Insert 20 on left

Inserting a Node in BST – Implementation

```
void IntBinaryTree::insertNode(int num) {
   TreeNode *newNode, *nodePtr;  // Pointer to create new node & traverse tree

   newNode = new TreeNode; // Create a new node
   newNode->value = num;
   newNode->left = newNode->right = NULL;

   if (!root)  root = newNode; // If tree is empty.
   else { // Tree is not empty
      nodePtr = root;
      while (nodePtr != NULL) {
         if (num < nodePtr->value) { // Left subtree
            if (nodePtr->left)  { nodePtr = nodePtr->left; }
            else { nodePtr->left = newNode; break;  }
         }
         else if (num > nodePtr->value) {  // Right subtree
            if (nodePtr->right)    nodePtr = nodePtr->right;
            else  {  nodePtr->right = newNode;   break; }
         }
         else {  cout << "Duplicate value found in tree.\n"; break; }
      }
   }
}
```

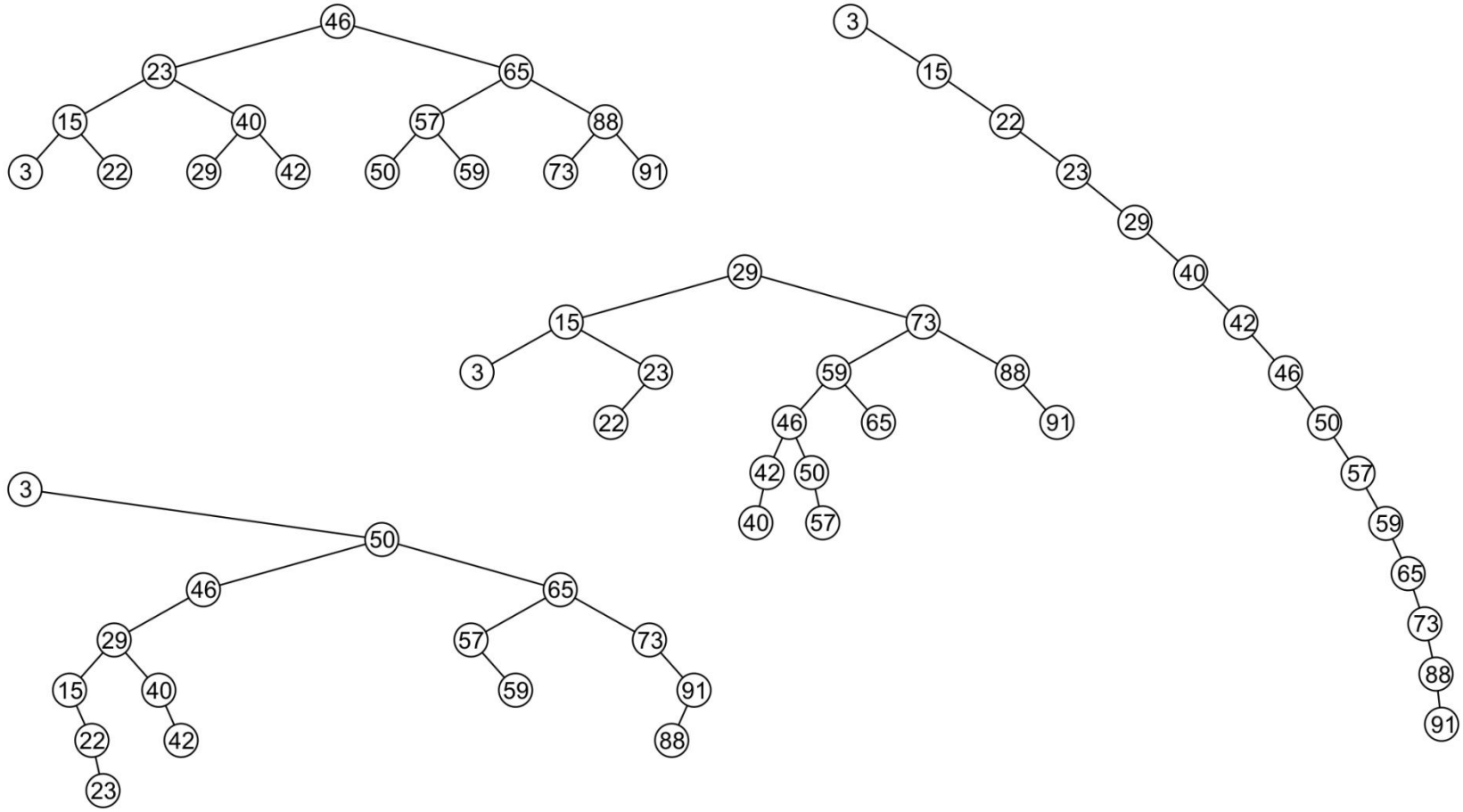
Inserting a Node in BST – Observation (1)

- Insertion may unbalance the tree
- It is possible to construct degenerate BST
  - The example is equivalent to a linked list



Inserting a Node in BST – Observation (2)

- All these binary search trees store the same data
  - Resultant tree depends on the order in which the values are inserted



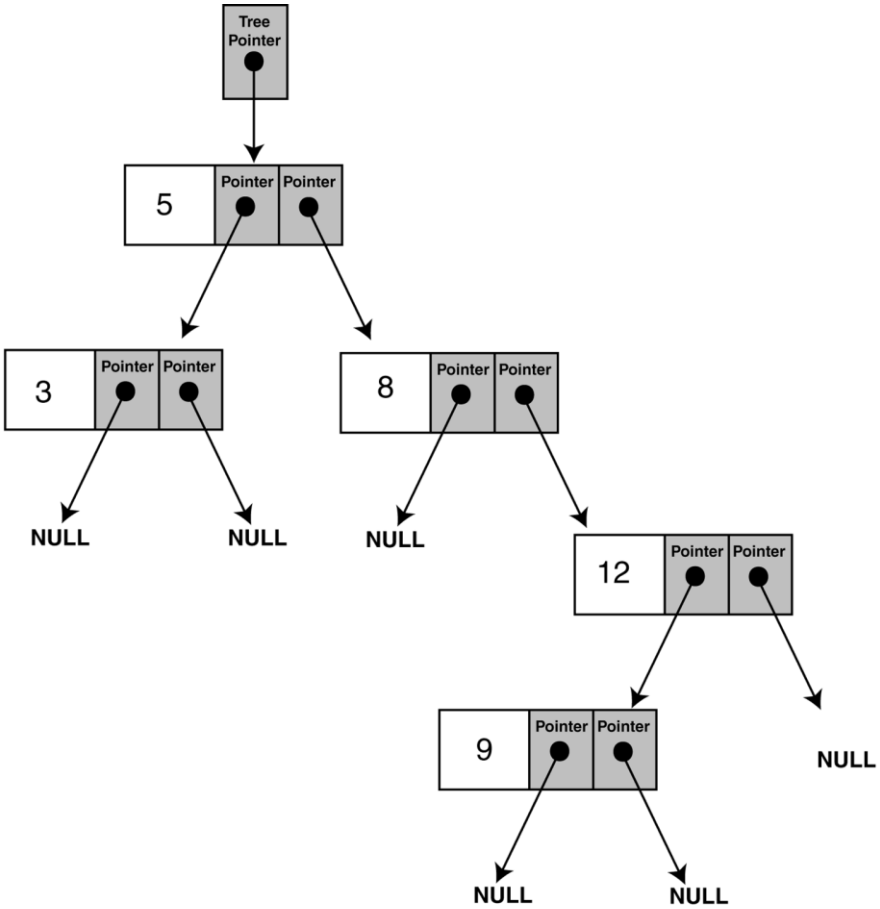
Using BST (1)

```
// This program builds a binary tree with 5 nodes.
// The SearchNode function determines if the
// value 3 is in the tree.
#include <iostream.h>
#include "IntBinaryTree.h“

void main(void)
{
    IntBinaryTree tree;

    cout << "Inserting nodes.\n";
    tree.insertNode(5);
    tree.insertNode(8);
    tree.insertNode(3);
    tree.insertNode(12);
    tree.insertNode(9);
    if (tree.find(3))
        cout << "3 is found in the tree.\n";
    else
        cout << "3 was not found in the tree.\n";
}
```

Output:
Inserting nodes.
3 is found in the tree.

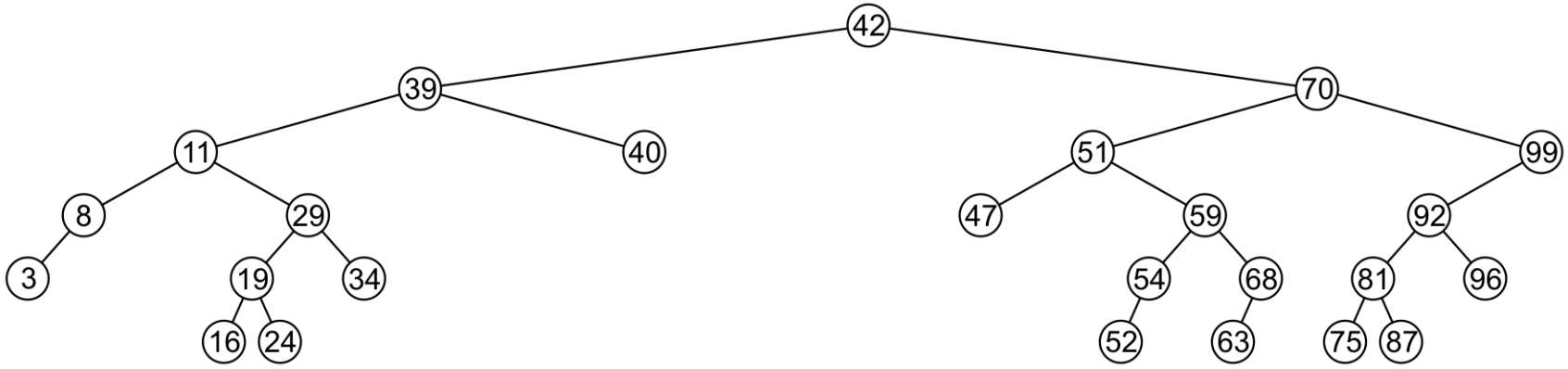
Using BST (2)

- Structure of binary tree built by the program



Deleting a Node

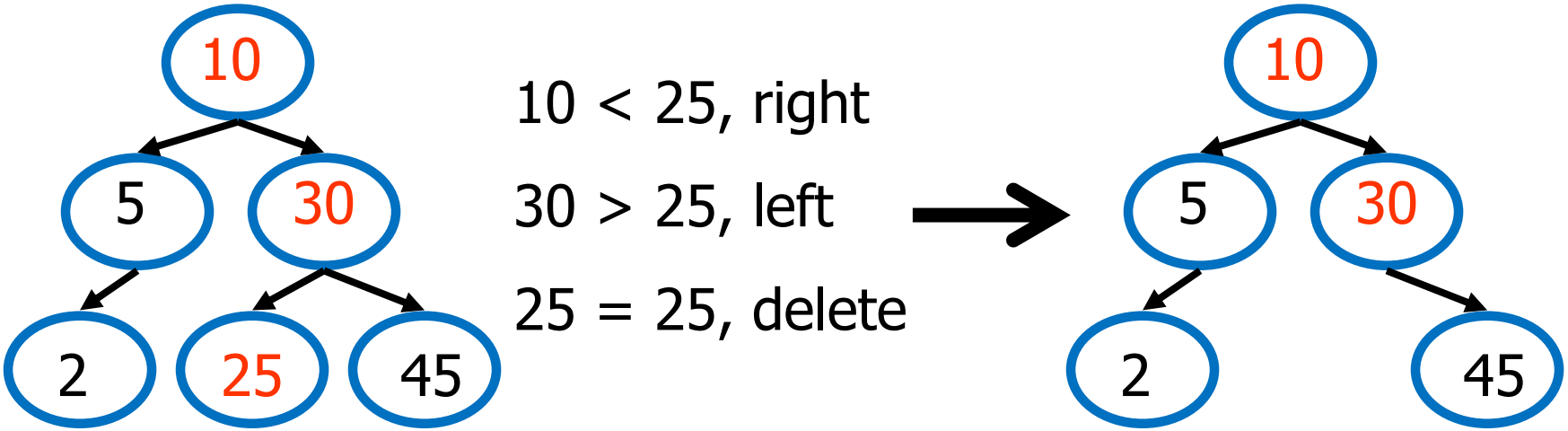
- A node being erased is not always going to be a leaf node
- There are three possible scenarios:
  - The node is a leaf node,
  - It has exactly one child, or
  - It has two children (it is a full node)



Deleting a Node – Leaf Node

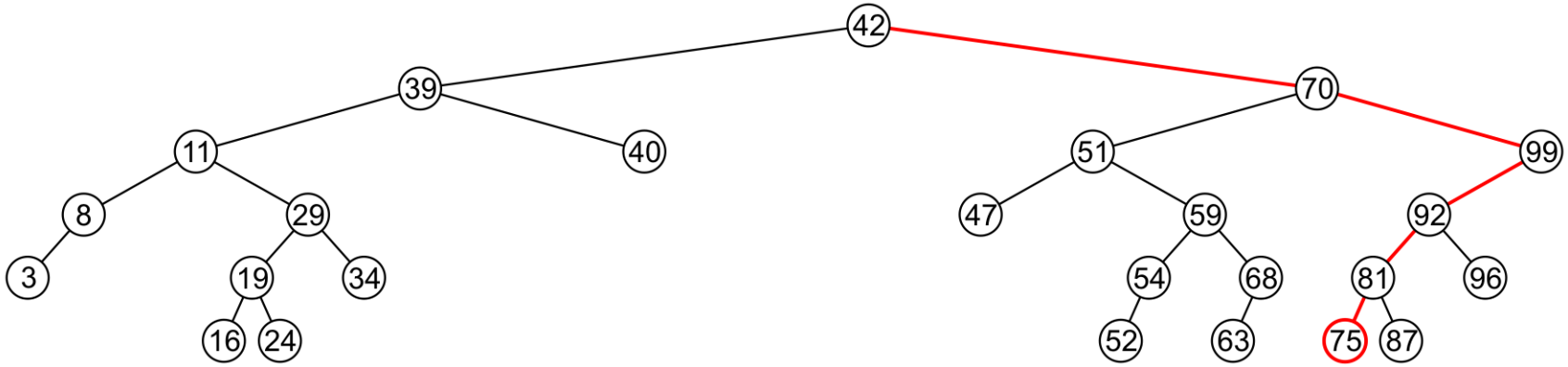
- Deleting a leaf node is easy
  - Find its parent
  - Set the child pointer that links to it to NULL
  - Free the node's memory

- Consider deleting node containing 25



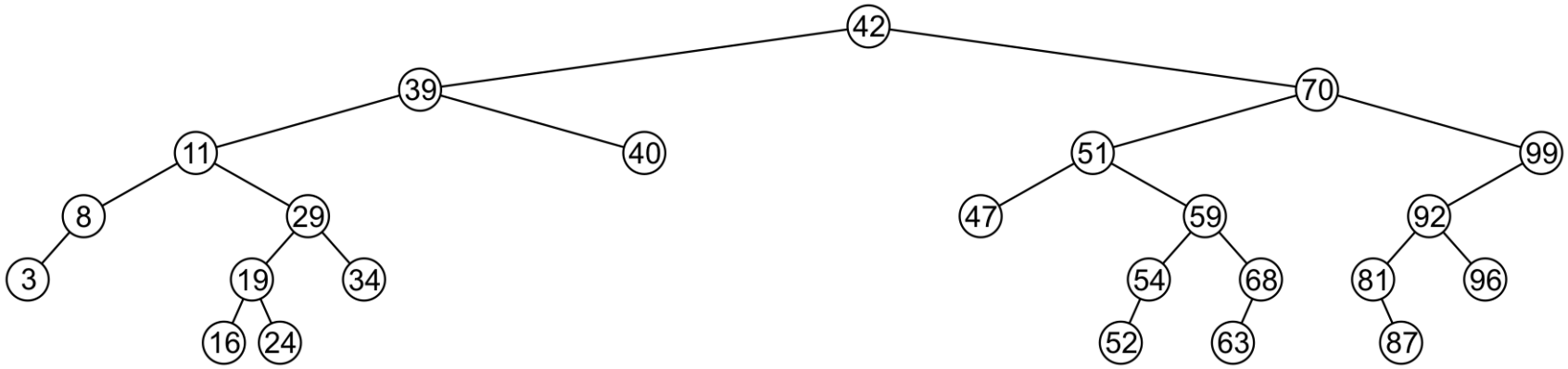
Deleting a Node – Leaf Node

- Consider deleting node containing 75



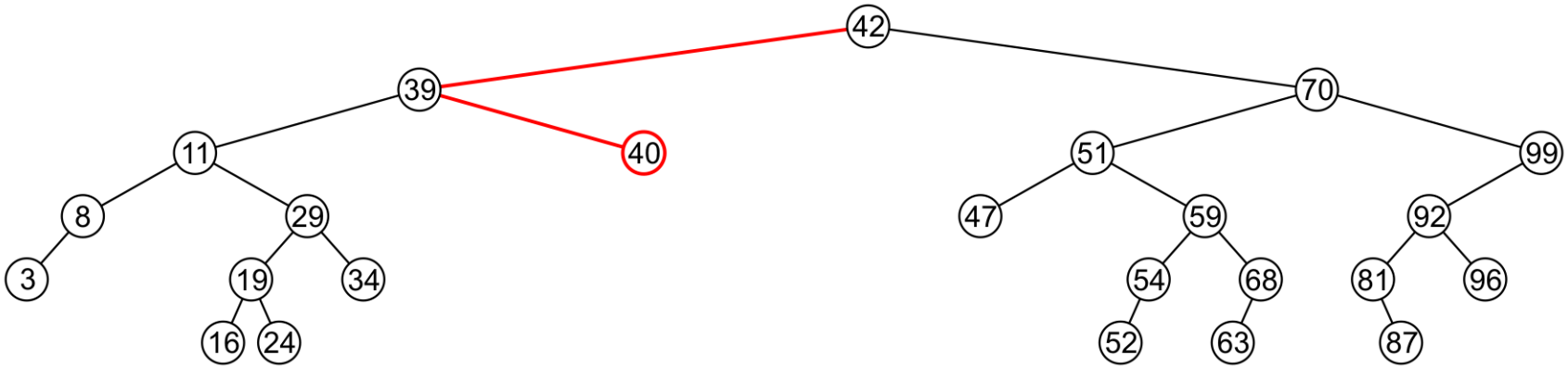
Deleting a Node – Leaf Node

- Consider deleting node containing 75
  - The node is deleted and left child of 81 is set to NULL



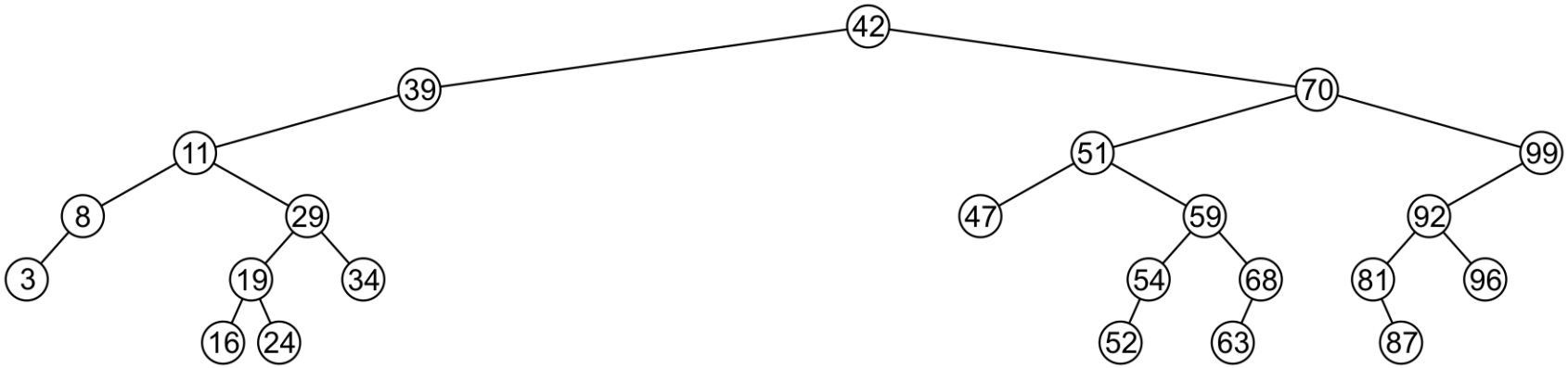
Deleting a Node – Leaf Node

- Consider deleting node containing 40



Deleting a Node – Leaf Node

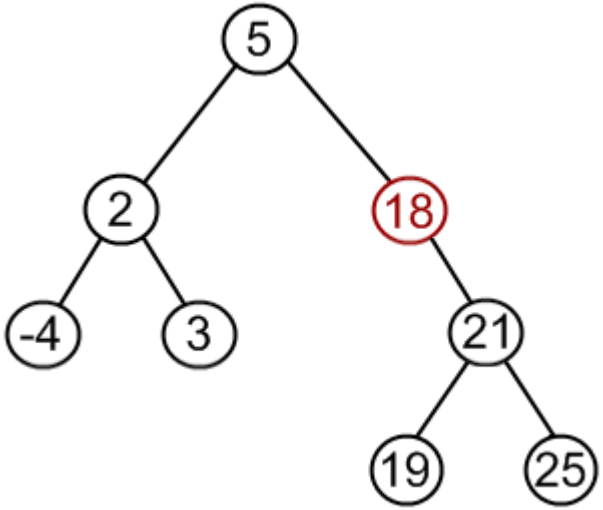
- Consider deleting node containing 40
  - Node is deleted and right child of 39 is set to NULL

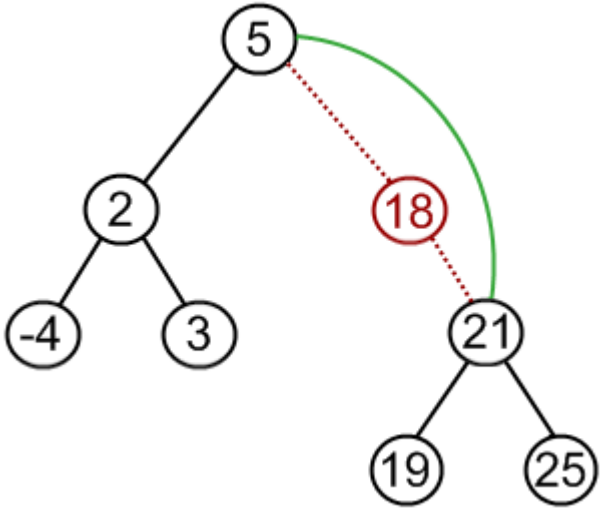


Deleting a Node – Node With Child

- If a node has only one child (left or right)
  - Simply promote the subtree associated with the child

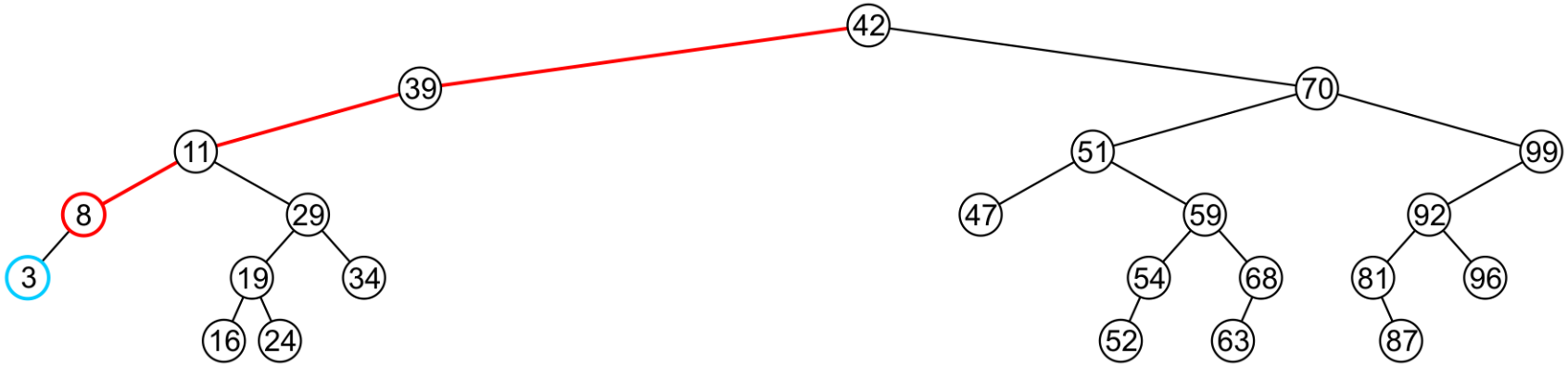
- Consider deleting 18 which has one right child
  - Node 18 is deleted and right tree of node 5 is update to point to 21





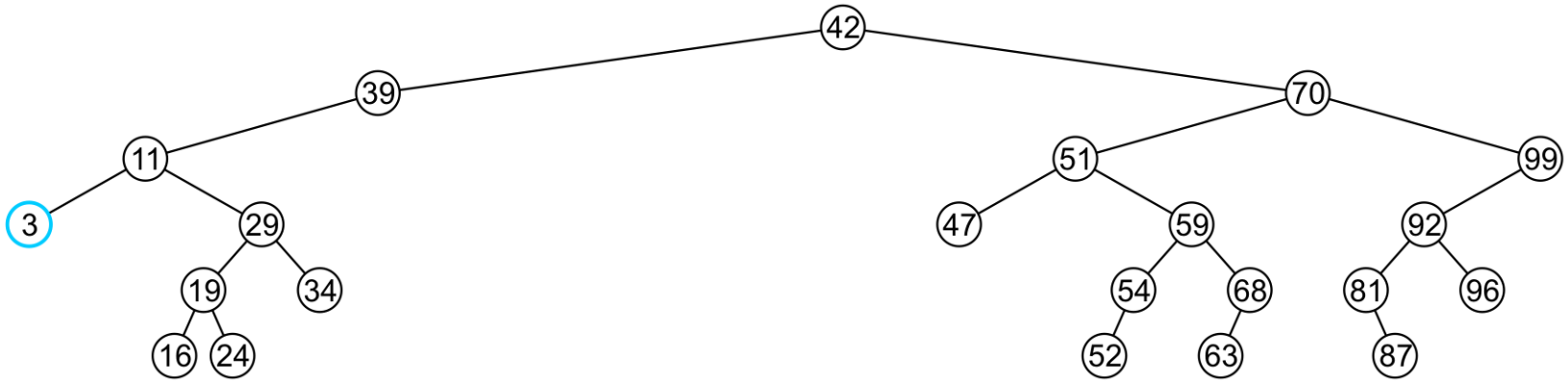
Deleting a Node – Node With Child

- Consider deleting 8 which has one left child



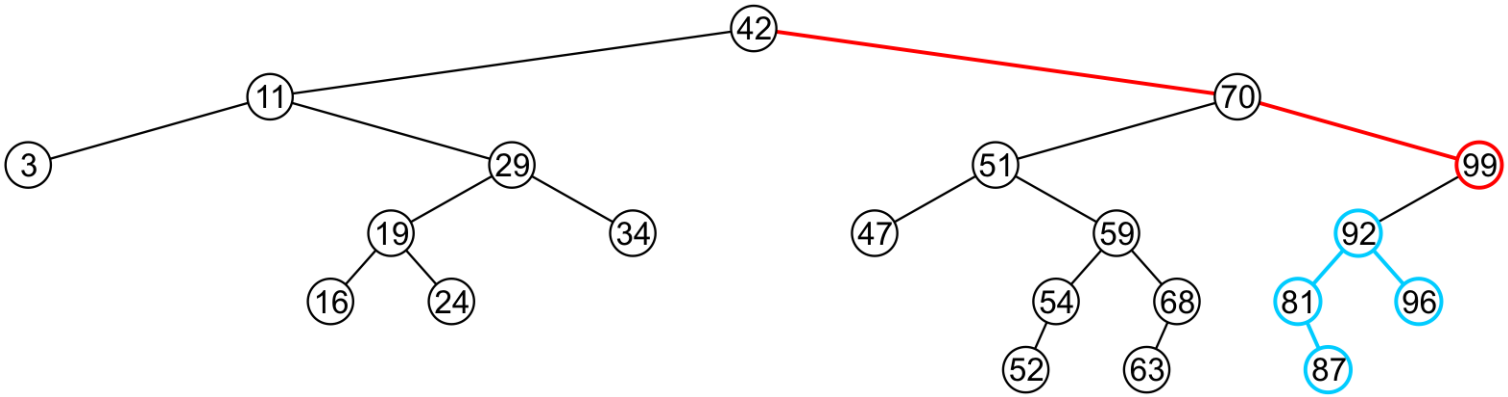
Deleting a Node – Node With Child

- Consider deleting 8 which has one left child
  - Node 8 is deleted and the left tree of 11 is updated to point to 3



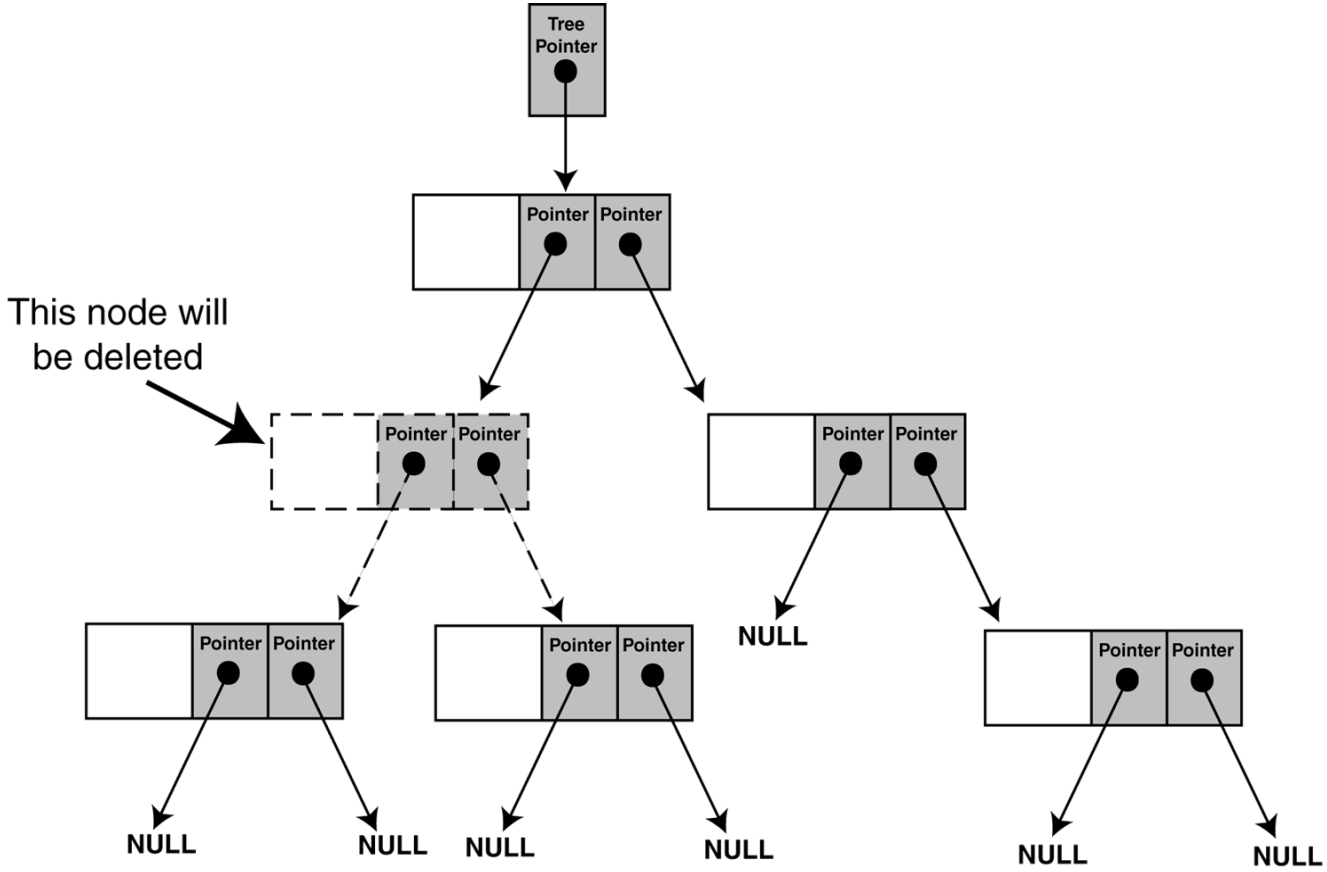
Deleting a Node – Node With Child

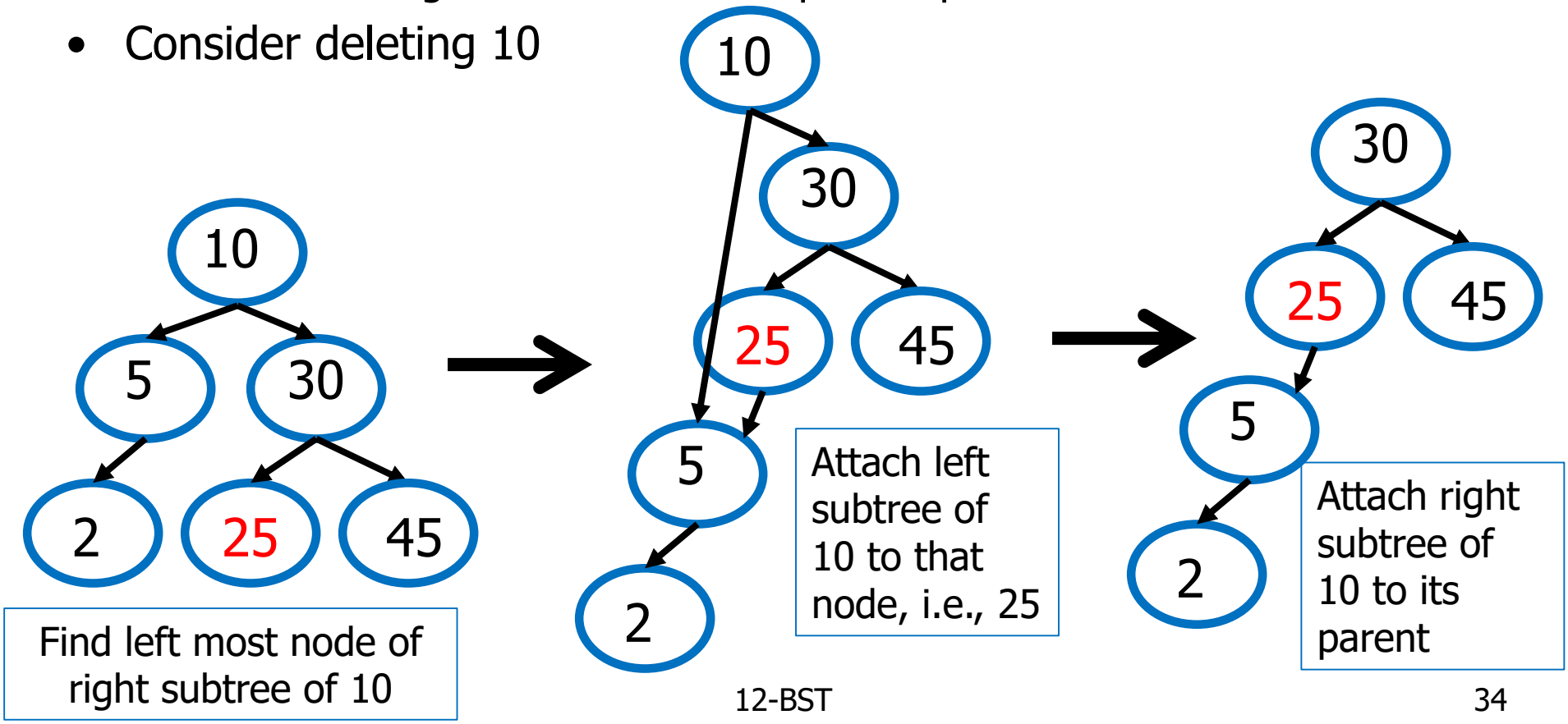
- Consider deleting the node containing 99



Deleting a Node – Node With Two Children

- The problem is not as easily solved if the node has two children



Deleting a Node – Node With Two Children

- Suppose node p with two children has to be deleted
  - Find a position in the right subtree of p to attach its left subtree
    - Left most node in the right subtree of node p
  - Attach the right subtree of node p to its parent
- Consider deleting 10